SERVERS

Some Basics

- Hardware sold for use as a server is qualitatively different from hardware sold for use as an individual workstation
  - Has different features and is engineered to a different economic model
  - Special procedures are used to install and support servers
- Typically have maintenance contracts, disk-backup systems, certified for use with a specific OS, better remote access, and servers reside in the controlled environment of a data center

Server Hardware for Servers

- Purchasing desktop hardware and loading it with server software may save you money but
  - May not good for long term
- Server hardware usually costs more but has additional features that justify the cost
- Features:
  - Extensibility
  - More CPU performance
  - High-performance I/O
  - Upgrade options
  - Rack mountable
  - No side access needs
  - High availability options
  - Maintenance contracts
  - Maintenance options

Enhancing Reliability with Servers

- Server appliance
- Redundant power supplies
- Full vs n+1 redundancy
- Hot swap components
- Multiple network connections
- Management Console

Server Processors

- Server processors have more cache memory than desktop processors
- Server processors can move / transfer more data at a time in the PCIe bus to other devices or network
- Server processor can run in multiple-CPU setup
- Server processor can also have more cores than desktop processors
- Server processors can also run ECC memory
  - ECC – error correcting code: ensures data sent to and from the RAM are transmitted correctly

Server Processors Available

- Intel Xeon Processors
  - D, W and E series
  - Scalable Processors
    - Platinum, Gold, Silver and Bronze series
- AMD Opteron and Epyc Processors
  - Opteron X3000 series
  - Epyc 7001 series
  - Epyc 7002 series

Alternative to Expensive Servers – Many Inexpensive Servers

- Use of commodity components
- Must be able to manage many servers
- Other strategies
  - Blade server
  - Virtual server

Types of Server

- Tower Servers
- Rack mount Servers
- Blade servers

Tower Servers

- Look very much like a desktop computer
- Hardware is different because it is more reliable and durable
  - Processors can vary much high-end desktop processors to entry level server processors
- Advantages
  - Airflow and cooling is good since components are not densely packed
  - Can use standard component sizes
- Disadvantage
  - Can be bulkier than rack-mount equivalent

Rack Mount Servers

- Can be mounted inside an equipment rack
- May run hotter than tower-based counter part
- May need special-sized components to fit inside chassis
- Server chassis can be 1u or 2u size
  - 2u size is better since it has better airflow

1u Server Chassis

2u Server Chassis

Blade Servers

- A server computer with modular design chassis allowing different number of servers and components to be interconnected inside the chassis
- Blade servers have a high-speed backplane to allow interconnection between servers and components
- Each server inside the chassis is treated as an individual server
- Other components inside the chassis can be
  - Network switch
  - Storage

An empty blade chassis

Blade Server - Front

8 blade servers followed by 6 empty slots

Captured from wikipedia

Blade Server - Back

- On the left two FC switches and two Ethernet switches
- On the right side a management module with console cables

Captured from wikipedia

Blade Server

- Advantage
  - Space saving since you can have multiple servers in a chassis
  - Less cables
  - Less total power consumption from the same number of rack servers
  - More CPUs in less space
- Disadvantage
  - Only blades and components can be installed from the same manufacturer

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